

Critical Multi-Angle Review Report

Version 32 - Polished Benchmarking Study

Manuscript Title:

A Benchmarking Study of Deep Learning Methods for Faint Comet Characterization:
Where Traditional Methods Fail and Physics-Informed Neural Networks Excel

Manuscript ID: APJ-2026-XXXX
Review Date: 2026 May 17
Review Type: **FINAL POLISH ASSESSMENT**
Recommendation: **ACCEPT with Minor Revisions**

Overall Score: 7.8/10 (Exceeds Acceptance Threshold)

Dimension	V31 Score	V32 Score	Change
Moral Integrity	9.5/10	9.5/10	→
Logical Consistency	7.0/10	7.5/10	↑
Practical Utility	7.0/10	7.5/10	↑
Data Authenticity	8.0/10	8.5/10	↑
Composite	7.5/10	7.8/10	Improved

Key Verdict

Version 32 represents the **culmination of iterative improvement**. The GitHub repository link is now actual (not placeholder), computational costs are fully disclosed, and the manuscript maintains complete transparency about its scope and limitations. Only the Zenodo DOI placeholder remains to be resolved before publication.

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1 Version Trajectory: A Story of Iterative Refinement

Version	Score	Recommendation	Trend	Key Characteristics
V1 (ddd006a)	4.8/10	Reject	–	Future citations, claimed real observations
V2 (f531082)	5.8/10	Major Revision	↑	Added “Simulation-Based” to title
V3 (16f6d3f)	6.5/10	Accept	↑	Removed future citations
V4 (1420fe3)	5.0/10	Major Revision	↓	REGRESSION: Reverted to observational
V31 (dff9aa5)	7.5/10	Accept/Very Minor	↑↑	“Benchmarking Study”, transparency
V32 (1c85a8b)	7.8/10	Accept/Minor	↑	Actual GitHub link, GPU specs

Version 32 Achievement: This version has crossed from “acceptable” to “polished.” The manuscript now meets the standards expected of high-impact astronomical methodology papers.

2 Executive Summary

2.1 Critical Assessment Summary

Version 32 maintains all the strengths of Version 31 while addressing several practical concerns:

- GitHub Repository Link Actualized:** Changed from placeholder [repository] to actual URL: `https://github.com/comet-dl/faint-comet-benchmark`
- Computational Cost Transparency:** Added detailed A100 GPU specifications and $105\times$ speedup quantification
- Complete Stress Test Suite:** All 5 stress tests now have proper titles, including Test 3: “Ultra-Low SNR (‘Whisper in a Hurricane’)”
- Uncertainty Budget Rigor:** Detailed decomposition into σ_{stat} , σ_{sys} , and δ_{domain}
- Zenodo DOI Still Placeholder:** Remains as [DOI] — the only remaining issue

2.2 Honesty and Scope Integrity: EXCELLENT

The manuscript continues to demonstrate exemplary scientific integrity:

- **Explicit “Not a Measurement Paper” Statement:** “We emphasize what this paper is not: it is not a measurement paper”
- **No Real 3I/ATLAS Claims:** Properly acknowledges 3I/ATLAS “lacks published multimodal observations”
- **N=1 Contextualization:** 2I/Borisov comparison explicitly labeled “feasibility demonstration, not statistical validation”
- **Synthetic Ground Truth:** All performance metrics against known synthetic data

3 Four-Dimension Critical Review

3.1 Dimension 1: Moral Integrity (9.5/10)

Assessment: **EXEMPLARY**

Criterion	Evaluation
Honesty	Full disclosure of synthetic-only validation; no inflated claims
Transparency	Complete failure boundary documentation; “Fail Gallery” included
Scope Clarity	Explicit statement of what the paper is AND is not
Attribution	Proper archival data acknowledgments (ESO, STScI)

Evidence:

“We emphasize what this paper is not: it is not a measurement paper. All performance metrics are evaluated against known synthetic ground truth... No physical parameters of any real comet are measured or claimed.”

Minor Note: Zenodo DOI placeholder needs actual DOI before publication.

3.2 Dimension 2: Logical Consistency (7.5/10)

Assessment: **STRONG**

Strengths:

- Stress test suite logically probes failure boundaries
- Uncertainty budget properly decomposed into three additive components
- If-Then guide directly derives from benchmark results

- PINN architecture specification complete (Table 1)

Improvement Over V31: Computational efficiency claims now backed by specific hardware (A100 GPU) and timing comparisons.

Task	Traditional	Proposed (A100)
Image reduction	38 min	1.5 min
Coma detection & masking	45 min manual	12 sec
Dust parameter inversion	6+ hr per epoch	2.1 min
Total per epoch	~7 hr	4 min

3.3 Dimension 3: Practical Utility (7.5/10)

Assessment: STRONG

The If-Then observational guide (Section 5.2) provides actionable decision rules:

1. **SNR > 5**, $|b| > 10^\circ$: Both methods reliable; advantage is computational
2. **SNR 2-5**, $|b| < 10^\circ$: Traditional methods biased; PINN preferred
3. **SNR < 2**: No automated method reliable; recommend stacking + limits
4. **No spectroscopy**: Gas parameters unconstrained ($2.7\times$ uncertainty)
5. **Extreme composition**: Retrain PINN before attempting inversion

Improvement: Guide now grounded in quantified uncertainty budget.

3.4 Dimension 4: Data Authenticity (8.5/10)

Assessment: VERY GOOD

Verification Checklist:

- ✓ All synthetic data generation parameters disclosed
- ✓ Random seeds specified: [42, 123, 456, 789, 1024]
- ✓ DIRTY code version/architecture documented
- ✓ Docker environment configuration available
- ✓ GitHub repository link actual and accessible
 - Zenodo DOI needs to be actual (not placeholder)

Uncertainty Budget Quantification:

$$\sigma_{\text{total}}^2 = \sigma_{\text{stat}}^2 + \sigma_{\text{sys}}^2 + \delta_{\text{domain}}^2$$

Dust: $\sigma_{\text{total}} \sim 8\%$

Gas: $\sigma_{\text{total}} \sim 12\%$

4 V31 vs V32: Detailed Comparison

Aspect	Version 31	Version 32
GitHub Link	Placeholder: [repository]	Actual: github.com/comet-dl/faint-comet-benchmark
Zenodo DOI	Placeholder: [DOI]	Placeholder: [DOI] (still needed)
Stress Test 3	Mentioned but title incomplete	Complete: “Ultra-Low SNR (‘Whisper in a Hurricane’)”
GPU Specs	Generic “GPU” mention	A100 detailed specs, 105× speedup
Processing Time	“4 min vs 7 hr” (no hardware)	Detailed breakdown with hardware context
Uncertainty	Qualitative discussion	Quantified: $\sigma_{\text{stat}}, \sigma_{\text{sys}}, \delta_{\text{domain}}$
Domain Error	Mentioned as “5-7%”	Derived from 2I/Borisov MAD decomposition

5 Required Revisions

5.1 Minor Revision (Must Fix)

Issue: Zenodo DOI Placeholder Location: Acknowledgments and Data Availability sections Current: https://zenodo.org/record/[DOI] Required: Replace with actual DOI before publication, e.g., https://zenodo.org/record/1234567 Note: This can be done at the proof stage if repository is not yet archived. Include note to production editor.

5.2 Optional Polish (Recommended)

1. **Training Cost Disclosure:** Consider adding total GPU-hours for full training (e.g., “48 V100-hours”) to aid reproducibility assessment
2. **DOI Placeholder Note:** Add parenthetical note: “(DOI to be assigned upon acceptance)”

6 Final Recommendation

ACCEPT with Minor Revisions

Version 32 is a **polished, honest, and scientifically rigorous** benchmarking study. The manuscript:

- ✓ Explicitly defines its scope and limitations
- ✓ Documents failure boundaries as carefully as successes
- ✓ Provides complete reproducibility information (seeds, code, environment)
- ✓ Quantifies uncertainties through proper error decomposition
- ✓ Offers practical guidance for observers

The only required change is replacing the Zenodo DOI placeholder with an actual DOI (or adding a note that it will be assigned post-acceptance).

Expected Final Score After Revision: 8.0–8.2/10

Reviewer Signature: APJ Stringent Standards Review Panel

Date: 2026 May 17